

# THEORY OF SUPPLY

## Section A

Answer **all** questions

**Duration 2hr 5mins**

1. Which is a risk-bearing economy of scale?

- A** greater bargaining power in purchasing from suppliers
- B** greater diversification of the product range
- C** lower costs in raising capital
- D** lower distribution costs by increasing market share

2. In many developed economies, clothes are designed by small firms and retailed by large firms.

What is the most likely explanation for this pattern?

|          | clothes design firms                                      | clothes retail firms                                   |
|----------|-----------------------------------------------------------|--------------------------------------------------------|
| <b>A</b> | need to be flexible to cope with frequent fashion changes | need to exploit marketing economies of scale           |
| <b>B</b> | need to employ highly specialised and skilled workers     | need to operate at a low minimum efficient scale       |
| <b>C</b> | need to operate at a high minimum efficient scale         | need to offer a wide range of products to survive      |
| <b>D</b> | need to overcome high barriers to entry into the industry | need to take advantage of technical economies of scale |

3. Which is an external economy of scale?

- A** cheaper costs from purchasing large quantities of inputs
- B** decreased interest rates on borrowed funds
- C** increased labour productivity
- D** relevant training facilities at a local college

4. What is the most likely reason that barriers to entry might lead to x-inefficiency?

- A The firm does not account for any external benefits.
- B The firm has no incentive to cut its costs of production.
- C The firm's price is set higher than its average costs.
- D The firm's price is set higher than its marginal costs.

5. A firm always operates with economies of scale in the long run.

What is the shape of its long-run average cost curve?

- A It is horizontal.
- B It is 'U' shaped.
- C It slopes downwards.
- D It slopes upwards.

6. As firm X grows in size, it specialises in a narrower range of products.

Which economies of scale will the firm be **less** able to benefit from?

- A financial
- B marketing
- C risk-bearing
- D technical

7. What is true about economies of scale but **not** diseconomies of scale?

- A They occur because average revenue is increasing.
- B They occur because of management policy changes.
- C They occur in the short run.
- D They occur with decreasing average cost.

8. Which is an internal economy of scale?
- A efficient local transport networks
  - B improved access to spare parts as a result of industry growth
  - C lower risks from supplying a wider range of customers
  - D the training of skilled labour at a college financed by local firms
9. What is **most** likely to rise as a firm expands?
- A energy costs per unit of output
  - B the cost of components
  - C the cost of monitoring workers
  - D the rate of interest paid on borrowing
10. What **cannot** be changed in the short run?
- A the level of stock held by firms
  - B the level of technology available
  - C the market price of goods
  - D the output of individual firms in an industry
11. What would explain an increase in the size of an industry and a fall in the average output of the firms in an industry?
- A an advance in communication methods which reduce coordination problems within firms
  - B an advance in technology which increases the cost advantage of capital-intensive methods of production
  - C an increase in barriers to entry into the industry combined with an increase in barriers to exit from the industry
  - D an increase in external economies of scale combined with a decrease in internal economies of scale

12. Which is an internal economy of scale?
- A** cheaper loans from a local bank resulting from a merger
  - B** efficient local transport networks
  - C** specialist training facilities provided by a local college
  - D** the reputation of an area for a particular product
13. Which is **not** a reason for the existence of small firms in a market dominated by large companies?
- A** Small firms can concentrate on providing specialist services.
  - B** Small firms enjoy greater economies of scale.
  - C** Small firms enter the market with new innovative ideas.
  - D** Small firms have good knowledge of local markets.
14. What are the correct formulae for calculating average cost (AC) and average revenue (AR)?

|          | AC              | AR              |
|----------|-----------------|-----------------|
| <b>A</b> | $\frac{TC}{MC}$ | $\frac{TR}{MR}$ |
| <b>B</b> | $\frac{TC}{P}$  | $P \times Q$    |
| <b>C</b> | $\frac{TC}{Q}$  | $\frac{TR}{Q}$  |
| <b>D</b> | $\frac{TC}{Q}$  | $\frac{TR}{P}$  |

key

MC = marginal cost

MR = marginal revenue

P = price

Q = quantity

TC = total cost

TR = total revenue

15. A firm has fixed costs of \$300 and can produce two units per hour. Its total variable costs are \$200 for one unit and \$300 for two units.

Which cost will fall by the lowest amount when the second unit is produced?

- A** average fixed cost
- B** average total cost
- C** average variable cost
- D** marginal cost

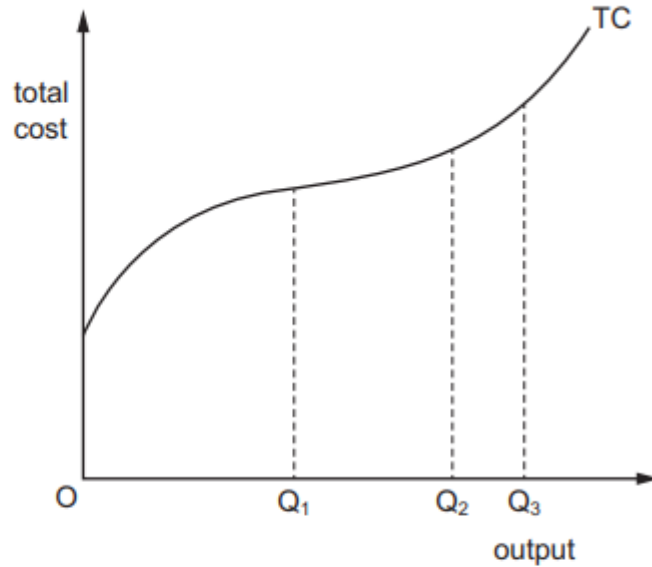
16. The table shows a firm's total and marginal costs.

| output | total cost (\$) | marginal cost (\$) |
|--------|-----------------|--------------------|
| 1      | 200             | 20                 |
| 2      | 215             | 15                 |
| 3      | 225             | 10                 |
| 4      | 240             | 15                 |
| 5      | 260             | 20                 |

What is the average fixed cost of producing 6 units?

- A** \$20      **B** \$30      **C** \$180      **D** \$200

17. The diagram shows a firm's short-run total cost curve (TC).



Which statement is correct?

- A** Average total cost is minimised at output  $OQ_2$ .
  - B** Average variable cost is minimised at output  $OQ_1$ .
  - C** Average variable cost is minimised at output  $OQ_3$ .
  - D** Marginal cost is minimised at output  $OQ_1$ .
18. To increase the number of cleaners at a local school from 10 to 11, the employer has to raise the hourly rate of pay from \$8.00 to \$8.50.

What is the marginal cost of labour per hour to the employer?

- A** \$0.50
- B** \$13.50
- C** \$88.50
- D** \$93.50

19. Diagram 1 shows the total cost curve for a firm. Diagram 2 shows possible marginal cost curves.



diagram 1

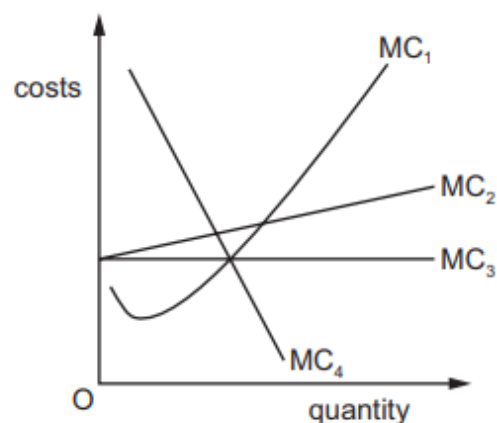
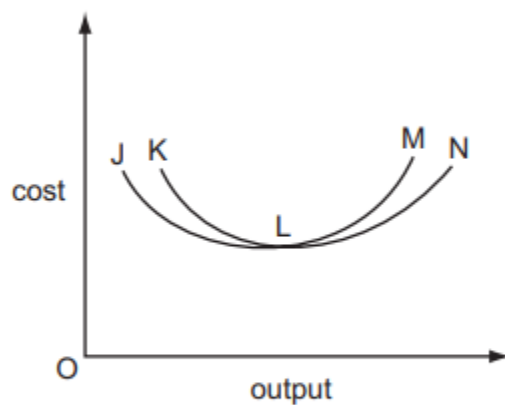


diagram 2

Which marginal cost curve is consistent with the total cost curve?

- A**  $MC_1$       **B**  $MC_2$       **C**  $MC_3$       **D**  $MC_4$

20. The diagram shows a firm's short-run and long-run average cost curves.



Which curve is the firm's long-run average cost curve?

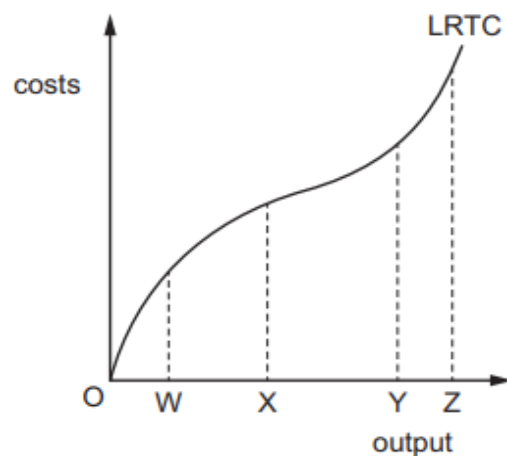
- A** JLM      **B** JLN      **C** KLM      **D** KLN

21. A firm employs a worker who adds less to output than the previous worker employed.

What does this illustrate?

- A** decreasing marginal costs
- B** diseconomies of scale
- C** increasing returns to scale
- D** the law of diminishing returns

22. The diagram shows the long-run total cost (LRTC) curve of a firm.



At which output is the long-run average total cost at its minimum?

- A** OW
- B** OX
- C** OY
- D** OZ

23. The following schedule gives information about a firm's costs.

|                      |        |
|----------------------|--------|
| total variable costs | \$6400 |
| average fixed costs  | \$20   |
| average total costs  | \$100  |

Which output would be consistent with these costs?

- A** 32 units
- B** 50 units
- C** 64 units
- D** 80 units

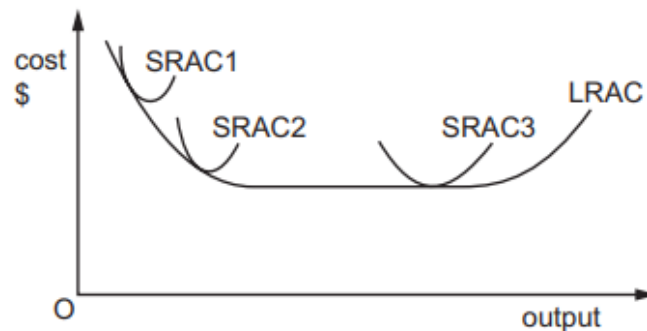
24. When output increases in the short run, which statement is correct?
- A** Average fixed cost first falls and then, beyond some point, rises.
  - B** Average variable cost increases as soon as the law of diminishing marginal returns begins to operate.
  - C** The minimum point of the average total cost curve occurs at a greater level of output than the minimum point of the average variable cost curve.
  - D** When marginal cost reaches its minimum point, average variable cost must be greater than average fixed cost.

25. A firm incurs both fixed and variable costs of production. It is currently producing at the level of output which minimises its average total cost.

What effect will a small reduction in output have on the firm's marginal and average variable costs?

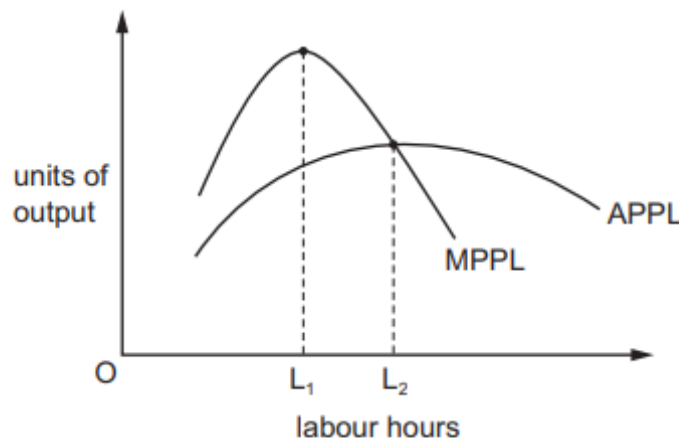
|          | marginal cost | average<br>variable cost |
|----------|---------------|--------------------------|
| <b>A</b> | fall          | fall                     |
| <b>B</b> | fall          | rise                     |
| <b>C</b> | rise          | fall                     |
| <b>D</b> | rise          | rise                     |

26. The diagram shows the long-run average cost (LRAC) curve for a firm, and three short-run average cost (SRAC) curves.



Which statement is correct?

- A When the minimum point of SRAC3 is lower than the minimum point of SRAC2 it reflects the law of variable proportions.
  - B The law of diminishing returns explains why each of the SRAC curves is U-shaped.
  - C The position of SRAC2 shows the minimum efficient scale of production.
  - D The rising section of SRAC1 indicates the existence of diseconomies of scale.
27. The diagram shows a firm's average physical product of labour and marginal physical product of labour curves.



What can be deduced from the diagram?

- A APPL declines beyond  $OL_2$  as a result of diseconomies of scale.
- B A profit-maximising firm would never hire more than  $OL_2$  labour hours.
- C Between  $OL_1$  and  $OL_2$ , each extra labour hour hired causes both MPPL and APPL to decline.
- D Diminishing returns first set in when the number of labour hours hired reaches  $OL_1$ .

28. The table shows the output of chairs at a factory when different numbers of workers are employed.

|                           |   |   |    |    |    |    |
|---------------------------|---|---|----|----|----|----|
| number of workers         | 0 | 1 | 2  | 3  | 4  | 5  |
| number of chairs produced | 0 | 7 | 17 | 26 | 34 | 40 |

When will diminishing marginal returns to labour set in?

- A** When the second worker is employed.
- B** When the third worker is employed.
- C** When the fourth worker is employed.
- D** When the fifth worker is employed.

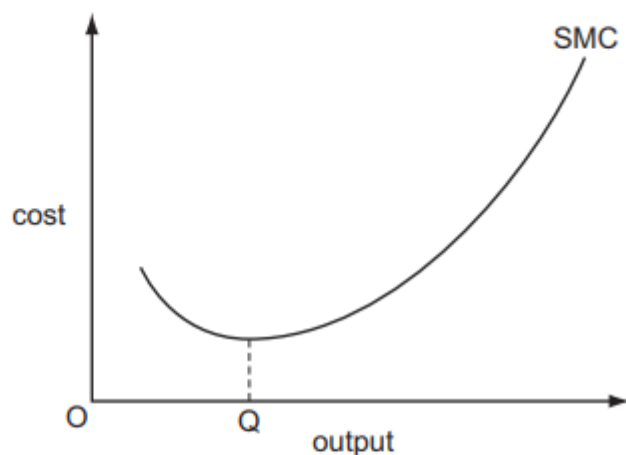
29. The table shows the total cost of a firm.

| output | total cost<br>\$ |
|--------|------------------|
| 0      | 40               |
| 1      | 55               |
| 2      | 60               |
| 3      | 65               |
| 4      | 80               |

What is the average variable cost of producing 4 units of output?

- A** \$10                      **B** \$15                      **C** \$20                      **D** \$40

30. The diagram shows a firm's short-run marginal cost curve.



What explains why the curve is upward sloping at output levels above OQ?

- A diseconomies of scale
  - B inelasticity of supply
  - C rising fixed costs
  - D the law of variable proportions
31. The table shows average total cost and marginal cost for a firm in perfect competition for a range of output.

| output                | 1  | 2   | 3 | 4    | 5   |
|-----------------------|----|-----|---|------|-----|
| average total cost \$ | 12 | 7.5 | 6 | 5.25 | 4.8 |
| marginal cost \$      | 3  | 3   | 3 | 3    | 3   |

What can be concluded from the above information?

- A The firm is operating with diseconomies of scale.
- B The firm's average variable cost is constant.
- C The firm's fixed costs are \$12.
- D The firm's revenue will not cover cost for this output.

32. When a firm doubles its variable inputs, its fixed inputs are unchanged and output less than doubles.

What does this illustrate?

- A decreasing average cost
  - B decreasing marginal cost
  - C diminishing returns
  - D diseconomies of scale
33. A firm operates at the maximum point on its average product curve.

What necessarily follows?

- A Average variable cost is at a minimum.
  - B Marginal cost is at a minimum.
  - C Marginal product is at a maximum.
  - D Total profit is at a maximum.
34. The table gives details of the carrying capacity and number of British licensed registered ships over 100 tons between 2010 and 2014.

| year                             | 2010 | 2011 | 2012 | 2013 | 2014 |
|----------------------------------|------|------|------|------|------|
| carrying capacity (million tons) | 17.3 | 16.6 | 17.7 | 17.6 | 15.2 |
| number of ships                  | 712  | 664  | 643  | 577  | 500  |

What can be concluded about British shipping between 2010 and 2014?

- A The average size of British registered ships increased.
- B The number of voyages made by British registered ships fell.
- C The productivity of British registered ships peaked in 2012.
- D The capacity of the British registered fleet fell continuously.

35. When a firm increases all its inputs by 100%, its output increases by 50%.

What does this illustrate?

- A decreasing returns to scale
- B economies of scale
- C the law of diminishing returns
- D the law of variable proportions

36. What is true about the average fixed costs of a firm?

- A They fall as output rises at first, but then rise.
- B They fall continuously as output rises.
- C They increase continuously as output rises.
- D They remain unchanged at all output levels.

37. Which statement describes the operation of the law of variable proportions?

- A As all factors are increased in proportion, the marginal cost of production increases.
- B As all factors are increased in proportion, total product rises at a diminishing rate.
- C As more of a variable factor is added to a fixed factor, the marginal cost of production diminishes.
- D As more of a variable factor is added to a fixed factor, total product rises at a diminishing rate.

38. According to the law of diminishing returns, what happens as more of a variable factor is combined with a fixed factor?

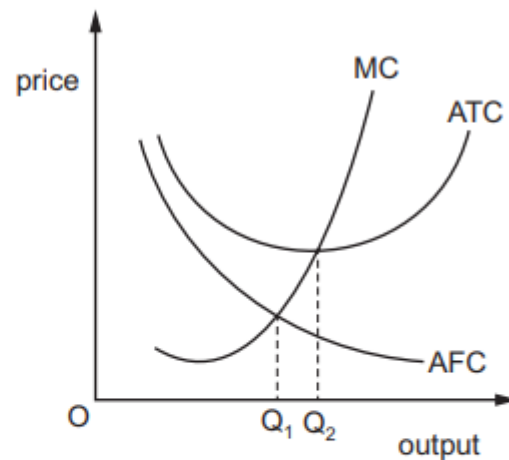
- A An increase in the price of the variable factor will eventually result in an increase in production costs.
- B A reduction in the quality of the variable factor will eventually result in an increase in production costs.
- C Fewer units of the variable factor will be needed to produce equal increases in output.
- D The proportions in which the factors are combined will eventually result in progressively smaller increases in output.

39. In 2015, a large mining company said it would reduce the number of staff by 6000 and sell its less profitable mines in an attempt to become more efficient.

If it is successful, what is most likely to happen to its costs?

|          | total fixed cost | total variable cost | average cost |
|----------|------------------|---------------------|--------------|
| <b>A</b> | fall             | fall                | fall         |
| <b>B</b> | fall             | fall                | no change    |
| <b>C</b> | fall             | no change           | fall         |
| <b>D</b> | no change        | fall                | fall         |

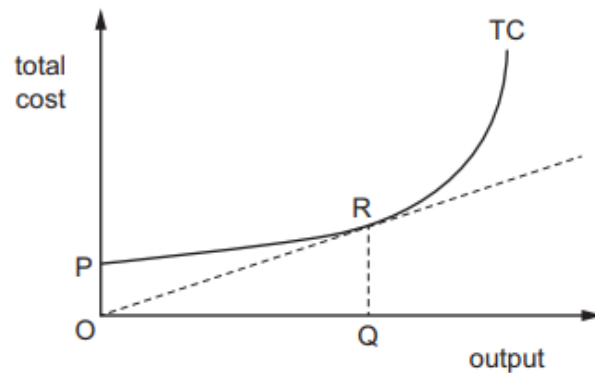
40. The diagram shows the cost curves for a firm.



What does the firm experience as it increases output from  $Q_1$  to  $Q_2$ ?

- A** decreased average variable cost
- B** diminishing returns
- C** economies of scale
- D** increased profit

41. In the diagram the curve TC shows the relationship between a firm's total costs and its level of output.

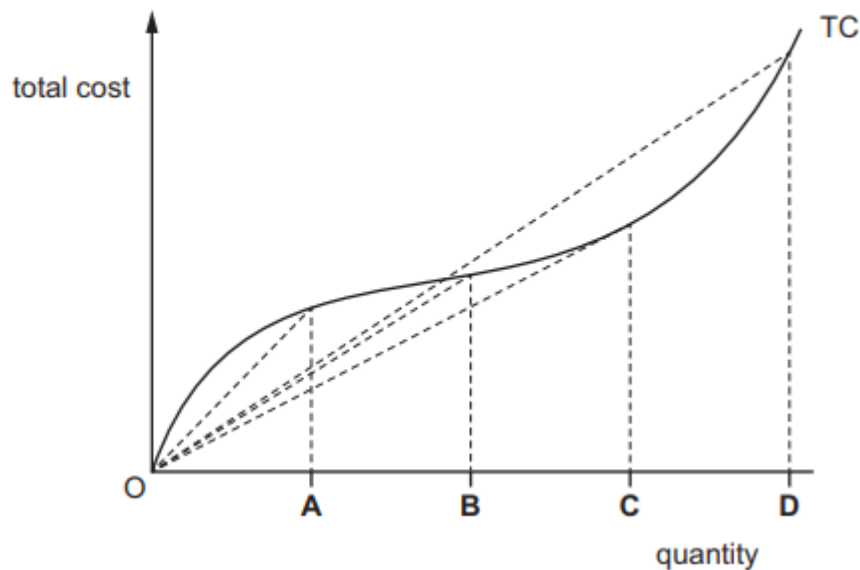


At output OQ average fixed costs are

- A equal to  $\frac{OP}{QR}$ .
  - B equal to  $\frac{QR}{OQ}$ .
  - C equal to  $\frac{OP}{OQ}$ .
  - D measured by the slope of the line OR.
42. A government imposes a \$10 tax on each unit produced of a product.
- How will this affect the firm?
- A Its average cost will rise by less than \$10.
  - B Its marginal cost will remain unchanged.
  - C Its minimum average cost output will be lower.
  - D Its minimum average cost output will be unchanged.

43. The diagram shows a firm's total cost curve.

At which level of output will the firm's marginal cost be equal to its average cost?



44. A café owner near a beach employed five staff and paid them each \$10 an hour. During the holiday season two extra staff were needed. To get extra staff the employer had to increase the hourly rate for all staff to \$12 an hour.

What was the marginal cost each hour of employing two extra staff?

- A \$2                      B \$24                      C \$34                      D \$84
45. In the year ending July 2016, airline fuel prices fell by 20.4% while the price of passenger aircraft increased by around 1.1%.

Assuming no other changes in the passenger airline industry, what was the outcome for fixed costs and variable costs?

|          | fixed costs | variable costs |
|----------|-------------|----------------|
| <b>A</b> | decreased   | decreased      |
| <b>B</b> | decreased   | increased      |
| <b>C</b> | increased   | decreased      |
| <b>D</b> | increased   | increased      |

46. The total cost to a school of employing ten cleaners is \$60 per hour.

The school estimates that the hourly marginal cost of employing an eleventh cleaner would be \$11.50.

By how much would the hourly wage have to increase to employ an additional cleaner?

- A** \$0.50                      **B** \$0.65                      **C** \$2.25                      **D** \$11.50

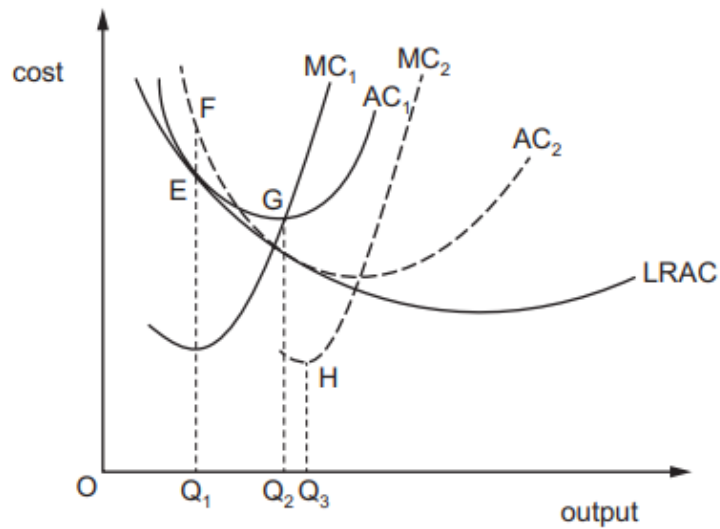
47. The table shows the total revenue and marginal cost of a firm at different levels of production.

| production<br>(tonnes) | total<br>revenue (\$) | marginal<br>cost (\$) |
|------------------------|-----------------------|-----------------------|
| 2                      | 70                    | 15                    |
| 3                      | 120                   |                       |
| 4                      | 160                   | 20                    |
| 5                      | 190                   | 25                    |
| 6                      | 220                   | 30                    |
| 7                      | 240                   | 40                    |

Within which output range will the firm's profits be maximised?

- A** 2–3 tonnes  
**B** 3–4 tonnes  
**C** 5–6 tonnes  
**D** 6–7 tonnes

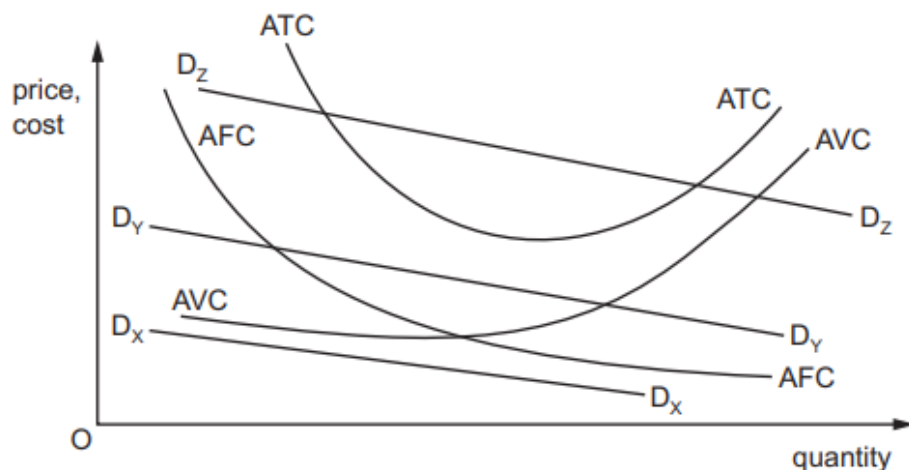
48. The diagram shows the short-run cost curves and the long-run average cost curve for a firm.



What can be concluded from the diagram?

- A** At output  $Q_1$ , point E represents a productively efficient position.
- B** At output  $Q_1$ , point F is preferred to point E because curve  $AC_2$  represents economies of scale.
- C** Point G at output  $Q_2$  is productively efficient in the long run.
- D** The biggest profits are made at point H at output  $Q_3$ , which is the lowest marginal cost position.

49. The diagram shows the average fixed cost (AFC), average variable cost (AVC) and average total cost (ATC) curves faced by three firms X, Y and Z.  $D_X$ ,  $D_Y$  and  $D_Z$  are the three respective demand curves. All three firms seek to make a profit.



Which statement is **not** correct?

- A Firm X will choose not to produce at all.
  - B Firm Y is likely to operate in the long run but not in the short run.
  - C Firm Y is likely to operate in the short run but not in the long run.
  - D Firm Z will operate in both the short run and the long run.
50. What is **not** a requirement for the law of diminishing returns to operate in the short run?
- A At least one factor will be fixed in quantity.
  - B Average returns to a variable factor will increase then fall.
  - C Marginal product of a variable factor will increase then fall.
  - D Inputs of all of the factors will be variable.

# THEORY OF SUPPLY (MARKING SCHEME)

# THEORY OF SUPPLY

## A LEVEL

### ECONOMICS P3

|     |   |     |   |     |   |      |   |      |   |
|-----|---|-----|---|-----|---|------|---|------|---|
| 1.  | B | 11. | D | 21. | D | 31.  | B | 41.  | C |
| 2.  | A | 12. | A | 22. | D | 32.  | C | 42.  | D |
| 3.  | D | 13. | B | 23. | C | 33.  | A | 43.  | C |
| 4.  | B | 14. | C | 24. | D | 34.  | A | 44.. | C |
| 5.  | C | 15. | C | 25. | C | 35.  | A | 45.  | C |
| 6.  | C | 16. | B | 26. | A | 36.  | B | 46.  | A |
| 7.  | D | 17. | D | 27. | A | 37.  | D | 47.  | C |
| 8.  | C | 18. | B | 28. | B | 38.. | D | 48.  | A |
| 9.  | C | 19. | C | 29. | A | 39.  | A | 49.  | B |
| 10. | B | 20. | B | 30. | D | 40.  | B | 50.  | D |